五 定 积 分

一、选择题

1.(B). 当 $0\leqslant x\leqslant 1$ 时，$F(x)=\int_0^x t^2\,dt=\frac{x^3}{3}$.

当 $1<x\leqslant 2$ 时，$F(x)=\int_0^x f(t)\,dt=\int_0^1 t^2\,dt+\int_1^x(2-t)\,dt$

$$=\frac{1}{3}+2x-2-\frac{x^2}{2}+\frac{1}{2}=-\frac{7}{6}+2x-\frac{x^2}{2}.$$

选(B).

2.(B). 当 $-1\leqslant x<0$ 时，$F(x)=\int_{-1}^x e^t\,dt=e^x-e^{-1}$；

当 $0\leqslant x\leqslant 1$ 时，$F(x)=\int_{-1}^0 e^t\,dt+\int_0^x t^2\,dt=1-e^{-1}+\frac{x^3}{3}$，即 $F(x)=\begin{cases}e^x-e^{-1}, & -1\leqslant x<0,\\ 1-e^{-1}+\frac{x^3}{3}, & 0\leqslant x\leqslant 1.\end{cases}$

因为 $F(0-0)=F(0)=F(0+0)=1-e^{-1}$，所以 $F(x)$ 在 $x=0$ 处连续.

$F'_-(0)=\lim\limits_{x\to 0^-}\frac{F(x)-F(0)}{x}=\lim\limits_{x\to 0^-}\frac{e^x-1}{x}=1$，$F'_+(0)=\lim\limits_{x\to 0^+}\frac{F(x)-F(0)}{x}=\lim\limits_{x\to 0^+}\frac{x^2}{3}=0$，

因为 $F'_-(0)\neq F'_+(0)$，所以 $F(x)$ 在 $x=0$ 处不可导. 选(B).

3.(C). $\sum\limits_{i=1}^{2n}\frac{i+2}{n^2}\left(1+\cos\frac{\pi i}{n}\right)^2=\frac{1}{n}\sum\limits_{i=1}^{2n}\frac{i}{n}\left(1+\cos\frac{\pi i}{n}\right)^2+\frac{2}{n^2}\sum\limits_{i=1}^{2n}\left(1+\cos\frac{\pi i}{n}\right)^2$.

由 $0\leqslant\frac{2}{n^2}\sum\limits_{i=1}^{2n}\left(1+\cos\frac{\pi i}{n}\right)^2\leqslant\frac{2}{n^2}\cdot 2n\cdot 4=\frac{16}{n}$，$\lim\limits_{n\to\infty}\frac{16}{n}=0$，得 $\lim\limits_{n\to\infty}\frac{2}{n^2}\sum\limits_{i=1}^{2n}\left(1+\cos\frac{\pi i}{n}\right)^2=0$，故

$$\text{原式}=\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{2n}\frac{i}{n}\left(1+\cos\frac{\pi i}{n}\right)^2=\int_0^2 x(1+\cos\pi x)^2\,dx$$

$$=\frac{1}{\pi^2}\int_0^2\pi x(1+\cos\pi x)^2\,d(\pi x)=\frac{1}{\pi^2}\int_0^{2\pi}x(1+\cos x)^2\,dx$$

$$\xlongequal{x-\pi=t}\frac{1}{\pi^2}\int_{-\pi}^{\pi}(\pi+t)(1-\cos t)^2\,dt=\frac{1}{\pi}\int_{-\pi}^{\pi}(1-\cos t)^2\,dt$$

$$=\frac{2}{\pi}\int_0^{\pi}(1-\cos t)^2\,dt=\frac{4}{\pi}\int_0^{\pi}\left(2\sin^2\frac{t}{2}\right)^2 d\left(\frac{t}{2}\right)$$

$$=\frac{16}{\pi}\int_0^{\frac{\pi}{2}}\sin^4 t\,dt=\frac{16}{\pi}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}=3.$$

选(C).

4.(B). 因为当 $0\leqslant x\leqslant 1$ 时，$\ln(1+x)\geqslant\frac{x}{1+x}\geqslant\frac{\sin x}{1+x}\geqslant\frac{\sin^2 x}{1+x}$，所以 $J<I$.

因为当 $0\leqslant x\leqslant 1$ 时，$0\leqslant\sin x\leqslant x\leqslant\tan x$，所以 $\ln(1+x)\leqslant\ln(1+\tan x)$，故 $I<K$，即 $J<I<K$. 选(B).

5.(D). $M=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{\sin x}{1+x^2}\cos^4 x\,dx=0$；

$N=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(\sin^3 x+\cos^4 x)\,dx=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^4 x\,dx>0$；

后续更新公众号[考机研]
永久联系微信4550060

$$P=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(x^2\sin^3x-\cos^4x)\mathrm{d}x=-\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^4x\,\mathrm{d}x<0.$$

故 $P<M<N$. 选(D).

6.(C). 因为 $f(t)-f(-t)$ 为奇函数,所以 $\int_a^x[f(t)-f(-t)]\mathrm{d}t$ 为偶函数. 选(C).

[注] 若 $f(t)$ 为偶函数,则 $\int_0^x f(t)\mathrm{d}t$ 为奇函数,但 $\int_a^x f(t)\mathrm{d}t$ 的奇偶性不确定.

7.(D). $I=\int_0^{uv}\frac{x}{u^2}f\left(v^2+\frac{x^2}{u^2}\right)\mathrm{d}x=\frac{1}{2}\int_0^{uv}f\left(v^2+\frac{x^2}{u^2}\right)\mathrm{d}\left(\frac{x^2}{u^2}\right)\overset{\frac{x^2}{u^2}=w}{=\!=\!=}\frac{1}{2}\int_0^{v^2}f(v^2+w)\mathrm{d}w$,

所以 I 只与 v 有关. 选(D).

8.(B). **方法一**

因为 $\lim\limits_{x\to0^+}(x-0)^1\cdot\frac{1}{\sin x}=1$ 且 $\alpha=1\geqslant1$,所以 $\int_0^2\frac{\mathrm{d}x}{\sin x}$ 发散,则 $\int_{-1}^2\frac{\mathrm{d}x}{\sin x}$ 发散;

因为 $\lim\limits_{x\to+\infty}x^1\cdot\frac{1}{\sqrt{x^2-1}}=1$ 且 $\alpha=1\leqslant1$,所以 $\int_1^{+\infty}\frac{\mathrm{d}x}{\sqrt{x^2-1}}$ 发散;

因为 $\lim\limits_{x\to0^+}(x-0)^1\cdot\frac{1}{x\sqrt{x+1}}=1$ 且 $\alpha=1\geqslant1$,所以 $\int_0^{+\infty}\frac{\mathrm{d}x}{x\sqrt{x+1}}$ 发散. 根据排除法,选(B).

方法二 $\int_0^{+\infty}\frac{\mathrm{d}x}{\sqrt{|x^3-x|}}=\int_0^1\frac{\mathrm{d}x}{\sqrt{x-x^3}}+\int_1^{+\infty}\frac{\mathrm{d}x}{\sqrt{x^3-x}}$.

由 $\lim\limits_{x\to0^+}(x-0)^{\frac{1}{2}}\cdot\frac{1}{\sqrt{x-x^3}}=\lim\limits_{x\to0^+}\frac{1}{\sqrt{1-x^2}}=1$ 且 $\alpha=\frac{1}{2}<1$;

$\lim\limits_{x\to1^-}(1-x)^{\frac{1}{2}}\cdot\frac{1}{\sqrt{x-x^3}}=\lim\limits_{x\to1^-}\frac{1}{\sqrt{x}\cdot\sqrt{1+x}}=\frac{1}{\sqrt{2}}$ 且 $\alpha=\frac{1}{2}<1$,得 $\int_0^1\frac{\mathrm{d}x}{\sqrt{x-x^3}}$ 收敛.

再由 $\lim\limits_{x\to1^+}(x-1)^{\frac{1}{2}}\cdot\frac{1}{\sqrt{x^3-x}}=\lim\limits_{x\to1^+}\frac{1}{\sqrt{x}\cdot\sqrt{x+1}}=\frac{1}{\sqrt{2}}$ 且 $\alpha=\frac{1}{2}<1$;

$\lim\limits_{x\to+\infty}x^{\frac{3}{2}}\cdot\frac{1}{\sqrt{x^3-x}}=1$ 且 $\alpha=\frac{3}{2}>1$,得 $\int_1^{+\infty}\frac{\mathrm{d}x}{\sqrt{x^3-x}}$ 收敛. 故 $\int_0^{+\infty}\frac{\mathrm{d}x}{\sqrt{|x^3-x|}}$ 收敛. 选(B).

二、填空题

1. 解
$$\begin{aligned}\lim_{n\to\infty}\sum_{i=1}^n\frac{i}{n^2}\left(1+\sin\frac{\pi i}{n}\right)^3&=\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^n\frac{i}{n}\left(1+\sin\frac{\pi i}{n}\right)^3=\int_0^1x(1+\sin\pi x)^3\mathrm{d}x\\&=\frac{1}{\pi^2}\int_0^1\pi x(1+\sin\pi x)^3\mathrm{d}(\pi x)=\frac{1}{\pi^2}\int_0^\pi x(1+\sin x)^3\mathrm{d}x\\&=\frac{1}{\pi^2}\cdot\frac{\pi}{2}\int_0^\pi(1+\sin x)^3\mathrm{d}x=\frac{1}{\pi}\int_0^{\frac{\pi}{2}}(1+\sin x)^3\mathrm{d}x\\&=\frac{1}{\pi}\int_0^{\frac{\pi}{2}}(1+3\sin x+3\sin^2x+\sin^3x)\mathrm{d}x=\frac{5}{4}+\frac{11}{3\pi}.\end{aligned}$$

2. 解 $x^2-\sin^2x=(x+\sin x)(x-\sin x)\sim2x\cdot\frac{x^3}{6}=\frac{x^4}{3}(x\to0)$,则

$$\text{原式}=\lim_{x\to0}\frac{\int_0^x\sqrt{1+t\cos t}\,\mathrm{d}t-\int_0^x\sqrt{1+\arcsin t}\,\mathrm{d}t}{\frac{x^4}{3}}=\frac{3}{4}\lim_{x\to0}\frac{\sqrt{1+x\cos x}-\sqrt{1+\arcsin x}}{x^3}$$

后续更新公众号[有机研]
永久联系微信4550060

$$=\frac{3}{4}\lim_{x\to 0}\frac{1}{\sqrt{1+x\cos x}+\sqrt{1+\arcsin x}}\cdot\frac{x\cos x-\arcsin x}{x^3}$$

$$=\frac{3}{8}\lim_{x\to 0}\frac{x\cos x-\arcsin x}{x^3}=\frac{3}{8}\left(\lim_{x\to 0}\frac{x\cos x-x}{x^3}-\lim_{x\to 0}\frac{\arcsin x-x}{x^3}\right)$$

$$=\frac{3}{8}\cdot\left[-\frac{1}{2}-\frac{1}{3}\lim_{x\to 0}\frac{(1-x^2)^{-\frac{1}{2}}-1}{x^2}\right]=-\frac{1}{4}.$$

3. 解 由 $\int_{\frac{1}{x}}^{1}f(xt)\,dt=\frac{1}{x}\int_{\frac{1}{x}}^{1}f(xt)\,d(xt)\xlongequal{xt=u}\frac{1}{x}\int_{1}^{x}f(u)\,du$，得

$$\lim_{x\to 1}\frac{\int_{\frac{1}{x}}^{1}f(xt)\,dt}{\ln^2 x}=\lim_{x\to 1}\frac{1}{x}\cdot\frac{\int_{1}^{x}f(u)\,du}{\ln^2[1+(x-1)]}=\lim_{x\to 1}\frac{\int_{1}^{x}f(u)\,du}{(x-1)^2}$$

$$=\frac{1}{2}\lim_{x\to 1}\frac{f(x)}{x-1}=\frac{1}{2}\lim_{x\to 1}\frac{f(x)-f(1)}{x-1}=\frac{1}{2}f'(1)=\frac{\pi}{2}.$$

4. 解 $\int_0^x tf(x^2-t^2)\,dt=-\frac{1}{2}\int_0^x f(x^2-t^2)\,d(x^2-t^2)\xlongequal{x^2-t^2=u}\frac{1}{2}\int_0^{x^2}f(u)\,du.$

由 $\lim_{x\to 0}\frac{x^2-\sin^2 x}{x^4}=\lim_{x\to 0}\frac{x+\sin x}{x}\cdot\frac{x-\sin x}{x^3}=\frac{2}{3}\lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{3}$，得 $x^2-\sin^2 x\sim\frac{1}{3}x^4$，从而

$$\lim_{x\to 0}\frac{\int_0^x tf(x^2-t^2)\,dt}{x^2-\sin^2 x}=\frac{3}{2}\lim_{x\to 0}\frac{\int_0^{x^2}f(u)\,du}{x^4}=\frac{3}{4}\lim_{x\to 0}\frac{f(x^2)}{x^2}=\frac{3}{2}.$$

5. 解 令 $\sqrt{e^{2x}-1}=t$，则

$$\int_0^{\ln 2}\sqrt{e^{2x}-1}\,dx=\int_0^{\sqrt{3}}t\cdot\frac{1}{2}\cdot\frac{2t}{1+t^2}dt=\int_0^{\sqrt{3}}\left(1-\frac{1}{1+t^2}\right)dt=\sqrt{3}-\frac{\pi}{3}.$$

6. 解 $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{|x|}{1-\sin x}dx=\int_0^{\frac{\pi}{4}}\left(\frac{x}{1-\sin x}+\frac{x}{1+\sin x}\right)dx=\int_0^{\frac{\pi}{4}}\frac{2x}{1-\sin^2 x}dx=2\int_0^{\frac{\pi}{4}}x\sec^2 x\,dx$

$$=2\int_0^{\frac{\pi}{4}}x\,d(\tan x)=2x\tan x\Big|_0^{\frac{\pi}{4}}-2\int_0^{\frac{\pi}{4}}\tan x\,dx$$

$$=\frac{\pi}{2}+2\ln\cos x\Big|_0^{\frac{\pi}{4}}=\frac{\pi}{2}-\ln 2.$$

7. 解 $\int_0^{\frac{\pi}{2}}\frac{dx}{\sin x+\cos x}=\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{2}}\frac{d\left(x-\frac{\pi}{4}\right)}{\cos\left(x-\frac{\pi}{4}\right)}\xlongequal{x-\frac{\pi}{4}=t}\frac{1}{\sqrt{2}}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec t\,dt$

$$=\sqrt{2}\int_0^{\frac{\pi}{4}}\sec t\,dt=\sqrt{2}\ln|\sec t+\tan t|\Big|_0^{\frac{\pi}{4}}=\sqrt{2}\ln(\sqrt{2}+1).$$

8. 解 $\int_{-\pi}^{\pi}\frac{x\sin x}{1+\cos^2 x}dx=2\int_0^{\pi}x\cdot\frac{\sin x}{1+\cos^2 x}dx=\pi\int_0^{\pi}\frac{\sin x}{1+\cos^2 x}dx$

$$=-\pi\int_0^{\pi}\frac{d(\cos x)}{1+\cos^2 x}=-\pi\arctan\cos x\Big|_0^{\pi}=\frac{\pi^2}{2}.$$

9. 解 $\int_0^1 f(x)\,dx=xf(x)\Big|_0^1-\int_0^1 x\sqrt{1+x^2}\,dx=-\frac{1}{2}\int_0^1(1+x^2)^{\frac{1}{2}}d(1+x^2)$

$$=-\frac{1}{3}(1+x^2)^{\frac{3}{2}}\Big|_0^1=\frac{1}{3}(1-2\sqrt{2}).$$

后续更新公众号[考机研]
永久联系微信4550060

10. 解 $\int_0^{e^2}\varphi(y)\mathrm{d}y=\varphi(y)\cdot y\Big|_0^{e^2}-\int_0^{e^2}y\mathrm{d}[\varphi(y)]=e\cdot e^2-\int_0^{e^2}y\mathrm{d}[\varphi(y)]=e^3-\int_1^e x^2\ln x\,\mathrm{d}x$

$=e^3-\frac{1}{3}\int_1^e\ln x\,\mathrm{d}(x^3)=e^3-\frac{1}{3}x^3\ln x\Big|_1^e+\frac{1}{3}\int_1^e x^2\mathrm{d}x=\frac{7e^3-1}{9}.$

［注］本题也可直接用第二类换元积分法.

三、解答题

1. 解 (1) 由 $0\leqslant\sum_{i=1}^n\frac{1}{n^3}\sqrt{1-\frac{i^2}{n^2}}\leqslant\sum_{i=1}^n\frac{1}{n^3}=\frac{1}{n^2}$，得 $\lim\limits_{n\to\infty}\sum_{i=1}^n\frac{1}{n^3}\sqrt{1-\frac{i^2}{n^2}}=0$，则

$$\lim_{n\to\infty}\sum_{i=1}^n\frac{i^2+1}{n^3}\sqrt{1-\frac{i^2}{n^2}}=\lim_{n\to\infty}\sum_{i=1}^n\frac{i^2}{n^3}\sqrt{1-\frac{i^2}{n^2}}=\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^n\frac{i^2}{n^2}\sqrt{1-\frac{i^2}{n^2}}=\int_0^1x^2\sqrt{1-x^2}\,\mathrm{d}x$$

$$\xlongequal{x=\sin t}\int_0^{\frac{\pi}{2}}\sin^2t\cdot\cos t\cdot\cos t\,\mathrm{d}t=\int_0^{\frac{\pi}{2}}\sin^2t(1-\sin^2t)\mathrm{d}t=I_2-I_4$$

$$=\frac{1}{2}\cdot\frac{\pi}{2}-\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}=\frac{\pi}{16}.$$

(2) $\frac{n^2}{n^2+1}\sum_{i=1}^n\frac{i}{n^2}\ln\left(1+\frac{2i^2}{n^2}\right)\leqslant\sum_{i=1}^n\frac{i}{n^2+\frac{1}{i}}\ln\left(1+\frac{2i^2}{n^2}\right)\leqslant\sum_{i=1}^n\frac{i}{n^2}\ln\left(1+\frac{2i^2}{n^2}\right).$

由 $\frac{n^2}{n^2+1}\to1(n\to\infty)$，得

$$\lim_{n\to\infty}\frac{n^2}{n^2+1}\sum_{i=1}^n\frac{i}{n^2}\ln\left(1+\frac{2i^2}{n^2}\right)=\lim_{n\to\infty}\sum_{i=1}^n\frac{i}{n^2}\ln\left(1+\frac{2i^2}{n^2}\right)=\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^n\frac{i}{n}\ln\left(1+\frac{2i^2}{n^2}\right)$$

$$=\int_0^1x\ln(1+2x^2)\mathrm{d}x=\frac{1}{4}\int_0^1\ln(1+2x^2)\mathrm{d}(1+2x^2)$$

$$=\frac{1}{4}\int_1^3\ln x\,\mathrm{d}x=\frac{1}{4}\left(x\ln x\Big|_1^3-2\right)=\frac{3\ln3-2}{4}.$$

由夹逼准则，得 $\lim\limits_{n\to\infty}\sum_{i=1}^n\frac{i}{n^2+\frac{1}{i}}\ln\left(1+\frac{2i^2}{n^2}\right)=\frac{3\ln3-2}{4}$.

2. 解 (1) $\int_0^2(x^2+2x)\sqrt{2x-x^2}\,\mathrm{d}x=\int_0^2[(x-1)^2+4(x-1)+3]\sqrt{1-(x-1)^2}\,\mathrm{d}(x-1)$

$$=\int_{-1}^1(x^2+4x+3)\sqrt{1-x^2}\,\mathrm{d}x$$

$$=\int_{-1}^1(x^2+3)\sqrt{1-x^2}\,\mathrm{d}x=2\int_0^1(x^2+3)\sqrt{1-x^2}\,\mathrm{d}x$$

$$\xlongequal{x=\sin t}2\int_0^{\frac{\pi}{2}}(\sin^2t+3)\cos t\cdot\cos t\,\mathrm{d}t$$

$$=2\int_0^{\frac{\pi}{2}}(\sin^2t+3)(1-\sin^2t)\mathrm{d}t=2\int_0^{\frac{\pi}{2}}(3-2\sin^2t-\sin^4t)\mathrm{d}t$$

$$=2\left(\frac{3\pi}{2}-2\cdot\frac{1}{2}\cdot\frac{\pi}{2}-\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}\right)=\frac{13\pi}{8}.$$

(2) $\int_0^1\frac{\mathrm{d}x}{x+\sqrt{1-x^2}}\xlongequal{x=\sin t}\int_0^{\frac{\pi}{2}}\frac{\cos t}{\sin t+\cos t}\mathrm{d}t=\int_0^{\frac{\pi}{2}}\frac{\cos x}{\sin x+\cos x}\mathrm{d}x=I.$

因为 $\int_0^{\frac{\pi}{2}}f(\sin x)\mathrm{d}x=\int_0^{\frac{\pi}{2}}f(\cos x)\mathrm{d}x$，所以 $I=\int_0^{\frac{\pi}{2}}\frac{\sin x}{\cos x+\sin x}\mathrm{d}x$，故

后续更新公众号[考研帮]
永久联系微信4550060

$$\text{原式}=\frac{1}{2}\left(\int_0^{\frac{\pi}{2}}\frac{\cos x}{\sin x+\cos x}\mathrm{d}x+\int_0^{\frac{\pi}{2}}\frac{\sin x}{\cos x+\sin x}\mathrm{d}x\right)=\frac{\pi}{4}.$$

$$(3)\int_{-1}^{1}\frac{x}{x+\sqrt{x^2+1}}\mathrm{d}x=\int_{-1}^{1}\frac{x(\sqrt{x^2+1}-x)}{(\sqrt{x^2+1}+x)(\sqrt{x^2+1}-x)}\mathrm{d}x$$

$$=\int_{-1}^{1}x(\sqrt{x^2+1}-x)\mathrm{d}x=-2\int_0^1x^2\mathrm{d}x=-\frac{2}{3}.$$

$$(4)\int_0^1\frac{\mathrm{d}x}{(1+x)\sqrt{1+x^2}}\overset{x=\tan t}{=\!=\!=}\int_0^{\frac{\pi}{4}}\frac{\sec^2t\,\mathrm{d}t}{(1+\tan t)\sec t}=\int_0^{\frac{\pi}{4}}\frac{\mathrm{d}t}{\sin t+\cos t}$$

$$=\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{4}}\sec\left(t-\frac{\pi}{4}\right)\mathrm{d}\left(t-\frac{\pi}{4}\right)$$

$$=\frac{1}{\sqrt{2}}\ln\left|\tan\left(t-\frac{\pi}{4}\right)+\sec\left(t-\frac{\pi}{4}\right)\right|\Big|_0^{\frac{\pi}{4}}=\frac{1}{\sqrt{2}}\ln(1+\sqrt{2}).$$

$$(5)\int_0^1x^n\sqrt{1-x^2}\,\mathrm{d}x\overset{x=\sin t}{=\!=\!=}\int_0^{\frac{\pi}{2}}\sin^nt\cdot\cos^2t\,\mathrm{d}t=\int_0^{\frac{\pi}{2}}\sin^nt\cdot(1-\sin^2t)\mathrm{d}t=I_n-I_{n+2}$$

$$=I_n-\frac{n+1}{n+2}I_n=\frac{1}{n+2}I_n.$$

当 $n=2k$ 时，原式 $=\dfrac{1}{2k+2}\cdot\dfrac{(2k-1)!!}{(2k)!!}\cdot\dfrac{\pi}{2}$；

当 $n=2k-1$ 时，原式 $=\dfrac{1}{2k+1}\cdot\dfrac{(2k-2)!!}{(2k-1)!!}$.

[注] $2k!!=2k\cdot(2k-2)\cdot\cdots\cdot4\cdot2$，$(2k-1)!!=(2k-1)\cdot(2k-3)\cdot\cdots\cdot3\cdot1$.

3. 解 $(1)\int_0^{\pi^2}(\sin\sqrt{x}+\cos^2\sqrt{x})\mathrm{d}x\overset{\sqrt{x}=t}{=\!=\!=}2\int_0^{\pi}t(\sin t+\cos^2t)\mathrm{d}t$

$$=2\cdot\frac{\pi}{2}\int_0^{\pi}(\sin t+\cos^2t)\mathrm{d}t=2\pi\int_0^{\frac{\pi}{2}}(\sin t+\cos^2t)\mathrm{d}t$$

$$=2\pi\left(1+\frac{1}{2}\cdot\frac{\pi}{2}\right)=2\pi\left(1+\frac{\pi}{4}\right).$$

(2) 因为 $\ln(x+\sqrt{1+x^2})$ 为奇函数，所以

$$\text{原式}=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^2x\cos^2x\,\mathrm{d}x=2\int_0^{\frac{\pi}{2}}\sin^2x(1-\sin^2x)\mathrm{d}x=2(I_2-I_4)$$

$$=2\left(\frac{1}{2}\cdot\frac{\pi}{2}-\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}\right)=\frac{\pi}{8}.$$

$$(3)\int_0^{\pi}\sqrt{\sin x-\sin^3x}\,\mathrm{d}x=2\int_0^{\frac{\pi}{2}}\sqrt{\sin x-\sin^3x}\,\mathrm{d}x=2\int_0^{\frac{\pi}{2}}\sqrt{\sin x}\cos x\,\mathrm{d}x$$

$$=2\int_0^{\frac{\pi}{2}}\sqrt{\sin x}\,\mathrm{d}(\sin x)=2\cdot\frac{2}{3}\sin^{\frac{3}{2}}x\Big|_0^{\frac{\pi}{2}}=\frac{4}{3}.$$

$$(4)\int_0^{\pi}x\sqrt{\sin^2x-\sin^4x}\,\mathrm{d}x=\frac{\pi}{2}\int_0^{\pi}\sqrt{\sin^2x-\sin^4x}\,\mathrm{d}x=\pi\int_0^{\frac{\pi}{2}}\sqrt{\sin^2x-\sin^4x}\,\mathrm{d}x$$

$$=\pi\int_0^{\frac{\pi}{2}}\sin x\cos x\,\mathrm{d}x=\pi\int_0^{\frac{\pi}{2}}\sin x\,\mathrm{d}(\sin x)=\frac{\pi}{2}\sin^2x\Big|_0^{\frac{\pi}{2}}=\frac{\pi}{2}.$$

$$(5)\int_{-\pi}^{\pi}|\sin x+\cos x|\,\mathrm{d}x=\sqrt{2}\int_{-\pi}^{\pi}\left|\sin\left(x+\frac{\pi}{4}\right)\right|\mathrm{d}\left(x+\frac{\pi}{4}\right)=\sqrt{2}\int_{-\frac{3\pi}{4}}^{\frac{5\pi}{4}}|\sin x|\,\mathrm{d}x.$$

因为 $f(x)=|\sin x|$ 的周期为 π，由周期函数的定积分性质，得

后续更新公众号[考研研]
永久联系微信4550060

$$原式=\sqrt{2}\int_0^{2\pi}|\sin x|\,dx=2\sqrt{2}\int_0^{\pi}\sin x\,dx=4\sqrt{2}\int_0^{\frac{\pi}{2}}\sin x\,dx=4\sqrt{2}.$$

4. 解 (1)$\int_0^1 3x^2\arcsin x\,dx\xrightarrow{x=\sin t}\int_0^{\frac{\pi}{2}}3t\sin^2 t\,d(\sin t)=\int_0^{\frac{\pi}{2}}t\,d(\sin^3 t)$

$$=t\sin^3 t\Big|_0^{\frac{\pi}{2}}-\int_0^{\frac{\pi}{2}}\sin^3 t\,dt=\frac{\pi}{2}-\frac{2}{3}.$$

(2)$\int_0^1 x^2\arctan x\,dx=\int_0^1\arctan x\,d\left(\frac{x^3}{3}\right)=\frac{x^3}{3}\arctan x\Big|_0^1-\frac{1}{3}\int_0^1\frac{x^3}{1+x^2}dx$

$$=\frac{\pi}{12}-\frac{1}{3}\int_0^1\left(x-\frac{x}{1+x^2}\right)dx=\frac{2\ln 2+\pi-2}{12}.$$

5. 解 (1)$\int_{\frac{1}{e}}^{e}\frac{|\ln x|}{x}dx=\int_{\frac{1}{e}}^{e}|\ln x|\,d(\ln x)=\int_{-1}^{1}|x|\,dx=2\int_0^1 x\,dx=1.$

(2)$\int_{\frac{1}{e}}^{1}\sqrt{(\ln x)^2}\,dx\xrightarrow{x=e^t}\int_{-1}^{0}|t|\cdot e^t\,dt=-\int_{-1}^{0}t\,d(e^t)$

$$=-te^t\Big|_{-1}^{0}+\int_{-1}^{0}e^t\,dt=-e^{-1}+1-e^{-1}=1-2e^{-1}.$$

6. 解 $\int_0^{2\pi}f(x-\pi)dx=\int_0^{2\pi}f(x-\pi)d(x-\pi)=\int_{-\pi}^{\pi}f(x)dx$

$$=\int_{-\pi}^{0}\frac{\sin x}{1+\cos^2 x}dx+\int_0^{\pi}x\sin^2 x\,dx=-\arctan(\cos x)\Big|_{-\pi}^{0}+\frac{\pi}{2}\int_0^{\pi}\sin^2 x\,dx$$

$$=-\frac{\pi}{2}+\frac{\pi}{2}\cdot 2\int_0^{\frac{\pi}{2}}\sin^2 x\,dx=-\frac{\pi}{2}+\frac{\pi}{2}\cdot 2\cdot\frac{1}{2}\cdot\frac{\pi}{2}=-\frac{\pi}{2}+\frac{\pi^2}{4}.$$

7. 解 $\int_0^{2\pi}f(x)dx=\int_0^{\pi}f(x)dx+\int_{\pi}^{2\pi}f(x)dx=\frac{\pi^2}{2}+\int_{\pi}^{2\pi}[f(x-\pi)+x\sin^2 x]dx$

$$=\frac{\pi^2}{2}+\int_{\pi}^{2\pi}f(x-\pi)d(x-\pi)+\int_{\pi}^{2\pi}x\sin^2 x\,dx$$

$$=\frac{\pi^2}{2}+\int_0^{\pi}f(x)dx+\int_{\pi}^{2\pi}x\sin^2 x\,dx\xrightarrow{x-\pi=t}\pi^2+\int_0^{\pi}(\pi+t)\sin^2 t\,dt$$

$$=\pi^2+\pi\int_0^{\pi}\sin^2 t\,dt+\int_0^{\pi}t\sin^2 t\,dt=\pi^2+2\pi\cdot\frac{1}{2}\cdot\frac{\pi}{2}+\frac{\pi}{2}\int_0^{\pi}\sin^2 t\,dt$$

$$=\frac{3\pi^2}{2}+\pi\cdot\frac{1}{2}\cdot\frac{\pi}{2}=\frac{3\pi^2}{2}+\frac{\pi^2}{4}=\frac{7\pi^2}{4}.$$

8. 解 令$\int_{-\pi}^{\pi}xf(x)dx=A$,则$f(x)=\sin^3 x+A$,$xf(x)=x\sin^3 x+Ax$,两边同时积分,得

$\int_{-\pi}^{\pi}xf(x)dx=\int_{-\pi}^{\pi}x\sin^3 x\,dx+\int_{-\pi}^{\pi}Ax\,dx$,即

$$A=\int_{-\pi}^{\pi}x\sin^3 x\,dx=2\int_0^{\pi}x\sin^3 x\,dx=\pi\int_0^{\pi}\sin^3 x\,dx=2\pi\int_0^{\frac{\pi}{2}}\sin^3 x\,dx=\frac{4\pi}{3},$$

从而$f(x)=\sin^3 x+\frac{4\pi}{3}$,故

$$\int_0^{\pi}f(x)dx=\int_0^{\pi}\left(\sin^3 x+\frac{4\pi}{3}\right)dx=\int_0^{\pi}\sin^3 x\,dx+\frac{4\pi}{3}\int_0^{\pi}dx=\frac{4}{3}(1+\pi^2).$$

9. 解 令$\int_{-\pi}^{\pi}f(x)\sin x\,dx=A$,则$f(x)=\frac{x}{1+\cos^2 x}+A$.

于是$f(x)\sin x=\frac{x\sin x}{1+\cos^2 x}+A\sin x$,两边同时从$-\pi$到$\pi$积分,得

后续更新公众号[考研研]
永久联系微信4550060

$$A=\int_{-\pi}^{\pi}\frac{x\sin x}{1+\cos^2 x}\mathrm{d}x=2\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}\mathrm{d}x=\pi\int_0^{\pi}\frac{\sin x}{1+\cos^2 x}\mathrm{d}x$$
$$=-\pi\arctan(\cos x)\Big|_0^{\pi}=\frac{\pi^2}{2},$$

则 $f(x)=\dfrac{x}{1+\cos^2 x}+\dfrac{\pi^2}{2}$.

10. 解 令 $A=\int_0^2 f(x)\mathrm{d}x$，$f(x)=x\sin^2\pi x+A\sqrt{2x-x^2}$，两边同时积分，得

$$A=\int_0^2 x\sin^2\pi x\,\mathrm{d}x+A\int_0^2\sqrt{2x-x^2}\,\mathrm{d}x.$$

而

$$\int_0^2 x\sin^2\pi x\,\mathrm{d}x=\frac{1}{\pi^2}\int_0^{2\pi}x\sin^2 x\,\mathrm{d}x\xlongequal{x-\pi=t}\frac{1}{\pi^2}\int_{-\pi}^{\pi}(\pi+t)\sin^2 t\,\mathrm{d}t=\frac{1}{\pi}\int_{-\pi}^{\pi}\sin^2 t\,\mathrm{d}t$$
$$=\frac{2}{\pi}\int_0^{\pi}\sin^2 t\,\mathrm{d}t=\frac{4}{\pi}\int_0^{\frac{\pi}{2}}\sin^2 t\,\mathrm{d}t=\frac{4}{\pi}\cdot\frac{1}{2}\cdot\frac{\pi}{2}=1,$$
$$\int_0^2\sqrt{2x-x^2}\,\mathrm{d}x=\int_0^2\sqrt{1-(x-1)^2}\,\mathrm{d}(x-1)=\int_{-1}^1\sqrt{1-x^2}\,\mathrm{d}x=2\int_0^1\sqrt{1-x^2}\,\mathrm{d}x$$
$$\xlongequal{x=\sin t}2\int_0^{\frac{\pi}{2}}\cos^2 t\,\mathrm{d}t=2\cdot\frac{1}{2}\cdot\frac{\pi}{2}=\frac{\pi}{2},$$

则 $A=1+\dfrac{\pi}{2}A$，解得 $A=\dfrac{2}{2-\pi}$，即 $\int_0^2 f(x)\mathrm{d}x=\dfrac{2}{2-\pi}$.

[注] $\int_{-1}^1\sqrt{1-x^2}\,\mathrm{d}x$ 不需要硬算，利用定积分的几何意义，可知 $\sqrt{1-x^2}$ 是半径为 1 的上半圆，面积为 $\dfrac{\pi}{2}$，即积分结果为 $\dfrac{\pi}{2}$.

11. 解 $\displaystyle\int_0^{\frac{\pi}{2}}\frac{f'(x)}{1+f^2(x)}\mathrm{d}x=\int_0^{\frac{\pi}{2}}\frac{1}{1+f^2(x)}\mathrm{d}[f(x)]=\arctan f(x)\Big|_0^{\frac{\pi}{2}}=\arctan f\left(\frac{\pi}{2}\right)$.

因为 $f\left(\dfrac{\pi}{2}\right)=\int_0^{\frac{\pi}{2}}\dfrac{\cos t}{1+\sin^2 t}\mathrm{d}t=\arctan(\sin t)\Big|_0^{\frac{\pi}{2}}=\dfrac{\pi}{4}$，所以原式 $=\arctan\dfrac{\pi}{4}$.

12. 解 $\displaystyle\int_0^1 x^2 f(x)\mathrm{d}x=\int_0^1 f(x)\mathrm{d}\left(\frac{x^3}{3}\right)=\frac{x^3}{3}f(x)\Big|_0^1-\frac{1}{3}\int_0^1 x^3 f'(x)\mathrm{d}x$

$$=-\frac{1}{3}\int_0^1 x^3\mathrm{e}^{-x^2}\mathrm{d}x=-\frac{1}{6}\int_0^1(-x^2)\mathrm{e}^{-x^2}\mathrm{d}(-x^2)$$
$$\xlongequal{-x^2=t}-\frac{1}{6}\int_0^{-1}t\mathrm{e}^t\mathrm{d}t=-\frac{1}{6}(t-1)\mathrm{e}^t\Big|_0^{-1}=\frac{1}{3\mathrm{e}}-\frac{1}{6}.$$

13. 解 $\displaystyle\int_0^1 f(x)\mathrm{d}x=f(1)-\int_0^1 x\arcsin(x-1)^2\mathrm{d}x=\int_0^1\arcsin(x-1)^2\mathrm{d}x-\int_0^1 x\arcsin(x-1)^2\mathrm{d}x$

$$=\int_0^1(1-x)\arcsin(x-1)^2\mathrm{d}x=-\frac{1}{2}\int_0^1\arcsin(x-1)^2\mathrm{d}(x-1)^2$$
$$\xlongequal{(x-1)^2=t}\frac{1}{2}\int_0^1\arcsin t\,\mathrm{d}t=\frac{1}{2}t\arcsin t\Big|_0^1-\frac{1}{2}\int_0^1\frac{t}{\sqrt{1-t^2}}\mathrm{d}t$$
$$=\frac{\pi}{4}+\frac{1}{2}\int_0^1\frac{-2t}{2\sqrt{1-t^2}}\mathrm{d}t=\frac{\pi}{4}+\frac{1}{2}\int_0^1\frac{1}{2\sqrt{1-t^2}}\mathrm{d}(1-t^2)$$
$$=\frac{\pi}{4}+\frac{1}{2}\sqrt{1-t^2}\Big|_0^1=\frac{\pi}{4}-\frac{1}{2}.$$

后续更新公众号[考机研]
永久联系微信4550060

14. 解 因为 $f(x)$ 为偶函数，所以只研究 $f(x)$ 在 $[0,+\infty)$ 内的最大值与最小值即可.

$f'(x)=2x(2-x^2)e^{-x^2}$，令 $f'(x)=0$，得 $f(x)$ 在 $[0,+\infty)$ 内的驻点为 $x=0, x=\sqrt{2}$.

当 $x\in(0,\sqrt{2})$ 时，$f'(x)>0$；当 $x\in(\sqrt{2},+\infty)$ 时，$f'(x)<0$.

所以 $x=\sqrt{2}$ 为函数 $f(x)$ 的极大值点，极大值为 $f(\sqrt{2})=1+\dfrac{1}{e^2}$.

因为 $f(0)=0, f(+\infty)=\int_0^{+\infty}(2-t)e^{-t}dt=2\Gamma(1)-\Gamma(2)=1$，所以最大值为 $1+\dfrac{1}{e^2}$，最小值为 0.

15. 解 (1) 因为 $\lim\limits_{x\to0^+}(x-0)^{\frac{1}{2}}\cdot\dfrac{\sqrt{-\ln x}}{x-1}=-\lim\limits_{x\to0^+}\sqrt{-x\ln x}=-\lim\limits_{x\to0^+}\sqrt{-\ln x^x}=0$ 且 $\alpha=\dfrac{1}{2}<1$，

又因为 $\lim\limits_{x\to1^-}(1-x)^{\frac{1}{2}}\cdot\dfrac{\sqrt{-\ln x}}{x-1}=-\lim\limits_{x\to1^-}\dfrac{\sqrt{-\ln[1+(x-1)]}}{\sqrt{1-x}}=-1$ 且 $\alpha=\dfrac{1}{2}<1$，所以

$\int_0^1\dfrac{\sqrt{-\ln x}}{x-1}dx$ 收敛.

(2) 因为 $\lim\limits_{x\to0^+}(x-0)^{\frac{1}{2}}\cdot\dfrac{1}{\sqrt{2x-x^2}}=\lim\limits_{x\to0^+}\dfrac{1}{\sqrt{2-x}}=\dfrac{1}{\sqrt{2}}$ 且 $\alpha=\dfrac{1}{2}<1$；

又因为 $\lim\limits_{x\to2^-}(2-x)^{\frac{1}{2}}\cdot\dfrac{1}{\sqrt{2x-x^2}}=\lim\limits_{x\to2^-}\dfrac{1}{\sqrt{x}}=\dfrac{1}{\sqrt{2}}$ 且 $\alpha=\dfrac{1}{2}<1$，所以 $\int_0^2\dfrac{dx}{\sqrt{2x-x^2}}$ 收敛.

16. 解 (1) $\int_0^{+\infty}\dfrac{x}{(1+x)^3}dx=\int_0^{+\infty}\left[\dfrac{1}{(1+x)^2}-\dfrac{1}{(1+x)^3}\right]dx$

$$=-\frac{1}{1+x}\Big|_0^{+\infty}+\frac{1}{2(1+x)^2}\Big|_0^{+\infty}=\frac{1}{2}.$$

(2) $\int_3^{+\infty}\dfrac{dx}{(x-1)^4\sqrt{x^2-2x}}=\int_3^{+\infty}\dfrac{d(x-1)}{(x-1)^4\sqrt{(x-1)^2-1}}\xlongequal{x-1=\sec t}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\dfrac{\sec t\tan t}{\sec^4 t\tan t}dt$

$$=\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}(1-\sin^2 t)\cos t\,dt=\frac{2}{3}-\frac{3\sqrt{3}}{8}.$$

(3) $\int_1^{+\infty}\dfrac{\arctan x}{x^2}dx=-\int_1^{+\infty}\arctan x\,d\left(\dfrac{1}{x}\right)=-\dfrac{\arctan x}{x}\Big|_1^{+\infty}+\int_1^{+\infty}\dfrac{1}{x(1+x^2)}dx$

$$=\frac{\pi}{4}+\frac{1}{2}\int_1^{+\infty}\left(\frac{1}{x^2}-\frac{1}{1+x^2}\right)d(x^2)=\frac{\pi}{4}+\frac{1}{2}\ln\frac{x^2}{1+x^2}\Big|_1^{+\infty}$$

$$=\frac{\pi}{4}+\frac{1}{2}\ln 2.$$

17. 证明 $S_1=(\xi-a)f(\xi), S_2=\int_\xi^b f(t)dt$，令 $\varphi(x)=(x-a)f(x)-\int_x^b f(t)dt$，显然

$$\varphi(x)=(x-a)f(x)+\int_b^x f(t)dt=\left[(x-a)\int_b^x f(t)dt\right]'.$$

再令 $h(x)=(x-a)\int_b^x f(t)dt$，因为 $h(a)=h(b)=0$，所以存在 $\xi\in(a,b)$，使得 $h'(\xi)=0$，即 $\varphi(\xi)=0$，可得 $(\xi-a)f(\xi)=\int_\xi^b f(t)dt$.

18. 证明 $\int_0^{\frac{\pi}{2}}\sin^n x\cos^n x\,dx=2^{-n-1}\int_0^{\frac{\pi}{2}}\sin^n 2x\,d(2x)=2^{-n-1}\int_0^{\pi}\sin^n x\,dx=2^{-n}\int_0^{\frac{\pi}{2}}\sin^n x\,dx$.

后续更新公众号[考机研]
永久联系微信4550060

19. 证明 $\int_0^{\frac{\pi}{2}} f(\sin x)\mathrm{d}x \xlongequal{x+t=\frac{\pi}{2}} \int_{\frac{\pi}{2}}^0 f(\cos t)(-\mathrm{d}t) = \int_0^{\frac{\pi}{2}} f(\cos t)\mathrm{d}t = \int_0^{\frac{\pi}{2}} f(\cos x)\mathrm{d}x.$

$I = \int_0^1 \frac{x^2}{x+\sqrt{1-x^2}}\mathrm{d}x \xlongequal{x=\sin t} \int_0^{\frac{\pi}{2}} \frac{\sin^2 t\cos t}{\sin t+\cos t}\mathrm{d}t$，由 $\int_0^{\frac{\pi}{2}} f(\sin x)\mathrm{d}x = \int_0^{\frac{\pi}{2}} f(\cos x)\mathrm{d}x$，得

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin^2 t\cos t}{\sin t+\cos t}\mathrm{d}t = \int_0^{\frac{\pi}{2}} \frac{\cos^2 t\sin t}{\cos t+\sin t}\mathrm{d}t,\text{则}$$

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sin^2 t\cos t}{\sin t+\cos t}\mathrm{d}t + \int_0^{\frac{\pi}{2}} \frac{\cos^2 t\sin t}{\cos t+\sin t}\mathrm{d}t = \int_0^{\frac{\pi}{2}} \sin t\cos t\,\mathrm{d}t$$

$$= \int_0^{\frac{\pi}{2}} \sin t\,\mathrm{d}(\sin t) = \frac{1}{2}\sin^2 t\Big|_0^{\frac{\pi}{2}} = \frac{1}{2},$$

故 $I = \int_0^1 \frac{x^2}{x+\sqrt{1-x^2}}\mathrm{d}x = \frac{1}{4}.$

20. (1) **解** $\int_a^b xf(x)f'(x)\mathrm{d}x = \frac{1}{2}\int_a^b x\,\mathrm{d}[f^2(x)] = \frac{x}{2}f^2(x)\Big|_a^b - \frac{1}{2}\int_a^b f^2(x)\mathrm{d}x = -\frac{1}{2}.$

(2) **证明** $\int_a^b xf(x)f'(x)\mathrm{d}x = -\frac{1}{2}$

$$\Rightarrow \left[\int_a^b xf(x)f'(x)\mathrm{d}x\right]^2 = \frac{1}{4} \leqslant \int_a^b f'^2(x)\mathrm{d}x\int_a^b x^2f^2(x)\mathrm{d}x.$$

21. 证明 **方法一** 令 $F(x) = \int_0^x f(t)\mathrm{d}t$，则 $F'(x) = f(x)$，于是 $\int_0^x \left[\int_0^t f(u)\mathrm{d}u\right]\mathrm{d}t = \int_0^x F(t)\mathrm{d}t$，

$$\int_0^x f(t)(x-t)\mathrm{d}t = x\int_0^x f(t)\mathrm{d}t - \int_0^x tf(t)\mathrm{d}t = xF(x) - \int_0^x t\,\mathrm{d}[F(t)]$$

$$= xF(x) - tF(t)\Big|_0^x + \int_0^x F(t)\mathrm{d}t = \int_0^x F(t)\mathrm{d}t.$$

方法二 因为 $\frac{\mathrm{d}}{\mathrm{d}x}\int_0^x \left[\int_0^t f(u)\mathrm{d}u\right]\mathrm{d}t = \int_0^x f(u)\mathrm{d}u,$

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_0^x f(t)(x-t)\mathrm{d}t = \frac{\mathrm{d}}{\mathrm{d}x}\left[x\int_0^x f(t)\mathrm{d}t - \int_0^x tf(t)\mathrm{d}t\right] = \int_0^x f(t)\mathrm{d}t,$$

所以 $\int_0^x \left[\int_0^t f(u)\mathrm{d}u\right]\mathrm{d}t - \int_0^x f(t)(x-t)\mathrm{d}t \equiv C_0$，取 $x=0$ 得 $C_0=0$，故

$$\int_0^x \left[\int_0^t f(u)\mathrm{d}u\right]\mathrm{d}t = \int_0^x f(t)(x-t)\mathrm{d}t.$$

22. 证明 (1) 令 $F(x) = \int_0^x f(t)\mathrm{d}t$，则 $F'(x) = f(x)$，$F(0) = F(1) = 0.$

由罗尔定理，存在 $c \in (0,1)$，使得 $F'(c) = 0$，即 $f(c) = 0.$

令 $h(x) = \mathrm{e}^x f(x)$，$h(0) = h(c) = h(1) = 0$，由罗尔定理，存在 $\xi_1 \in (0,c)$，$\xi_2 \in (c,1)$，使得 $h'(\xi_1) = h'(\xi_2) = 0$，而 $h'(x) = \mathrm{e}^x[f'(x) + f(x)]$，由 $\mathrm{e}^x \neq 0$，得

$$f'(\xi_1) + f(\xi_1) = 0,\ f'(\xi_2) + f(\xi_2) = 0.$$

(2) 令 $\varphi(x) = \mathrm{e}^{-x}[f'(x)+f(x)]$，则 $\varphi(\xi_1) = \varphi(\xi_2) = 0$，由罗尔定理，存在 $\xi \in (\xi_1,\xi_2) \subset (0,1)$，使得 $\varphi'(\xi) = 0$，而 $\varphi'(x) = \mathrm{e}^{-x}[f''(x) - f(x)]$，由 $\mathrm{e}^{-x} \neq 0$，得 $f''(\xi) = f(\xi).$

23. 证明 $f(x) = f(x) - f(0) = f'(\xi)x\ (0 < \xi < x)$，则

$$\left|\int_0^1 f(x)\mathrm{d}x\right| \leqslant \int_0^1 |f(x)|\mathrm{d}x = \int_0^1 |f'(\xi)|x\,\mathrm{d}x \leqslant \int_0^1 x\,\mathrm{d}x = \frac{1}{2}.$$

后续更新公众号[考机研]
永久联系微信4550060

24. 证明 显然 $f(x)=f(x)-f(0)=\int_0^x f'(t)\mathrm{d}t$，则

$$f^2(x)=\left[\int_0^x f'(t)\mathrm{d}t\right]^2=\left[\int_0^x 1\cdot f'(t)\mathrm{d}t\right]^2\leqslant\int_0^x 1^2\mathrm{d}t\cdot\int_0^x f'^2(t)\mathrm{d}t$$
$$=x\int_0^x f'^2(t)\mathrm{d}t\leqslant x\int_0^1 f'^2(t)\mathrm{d}t=x\int_0^1 f'^2(x)\mathrm{d}x,$$

故 $\int_0^1 f^2(x)\mathrm{d}x\leqslant\int_0^1 x\mathrm{d}x\cdot\int_0^1 f'^2(x)\mathrm{d}x=\frac{1}{2}\int_0^1 f'^2(x)\mathrm{d}x$.

25. 证明 令 $\varphi(x)=x^2f(x)$.

由积分中值定理，得 $f(1)=2\int_0^{\frac{1}{2}}x^2f(x)\mathrm{d}x=c^2f(c)$，其中 $c\in\left[0,\frac{1}{2}\right]$，即 $\varphi(c)=\varphi(1)$.

显然 $\varphi(x)$ 在区间 $[0,1]$ 上可导，由罗尔定理知，存在 $\xi\in(c,1)\subset(0,1)$，使得 $\varphi'(\xi)=0$.

而 $\varphi'(x)=2xf(x)+x^2f'(x)$，所以 $2\xi f(\xi)+\xi^2f'(\xi)=0$.

注意到 $\xi\neq0$，故 $2f(\xi)+\xi f'(\xi)=0$.

26. 分析 由 $f(x)\int_x^b g(t)\mathrm{d}t=g(x)\int_a^x f(t)\mathrm{d}t$，得 $g(x)\int_a^x f(t)\mathrm{d}t+f(x)\int_b^x g(t)\mathrm{d}t=0$，即

$$\left[\int_a^x f(t)\mathrm{d}t\int_b^x g(t)\mathrm{d}t\right]'=0,$$

则辅助函数为 $\varphi(x)=\int_a^x f(t)\mathrm{d}t\int_b^x g(t)\mathrm{d}t$.

证明 令 $\varphi(x)=\int_a^x f(t)\mathrm{d}t\int_b^x g(t)\mathrm{d}t$，显然 $\varphi(x)$ 在 $[a,b]$ 上可导. 又由于 $\varphi(a)=\varphi(b)=0$，由罗尔定理知，存在 $\xi\in(a,b)$，使得 $\varphi'(\xi)=0$，而 $\varphi'(x)=f(x)\int_b^x g(t)\mathrm{d}t+g(x)\int_a^x f(t)\mathrm{d}t$，所以 $f(\xi)\int_b^\xi g(x)\mathrm{d}x+g(\xi)\int_a^\xi f(x)\mathrm{d}x=0$，即 $f(\xi)\int_\xi^b g(x)\mathrm{d}x=g(\xi)\int_a^\xi f(x)\mathrm{d}x$.

27. 证明 由泰勒公式得 $f(x)=f(1)+f'(1)(x-1)+\frac{f''(\xi)}{2!}(x-1)^2$，其中 ξ 介于 1 与 x 之间，因为 $f''(x)>0$，所以 $f(x)\geqslant f(1)+f'(1)(x-1)$，当且仅当 $x=1$ 时等号成立，故

$$\int_0^2 f(x)\mathrm{d}x>2f(1)+\int_0^2 f'(1)(x-1)\mathrm{d}x=2f(1).$$

后续更新公众号[有机研]
永久联系微信4550060